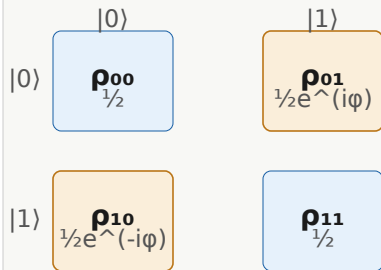


$\rho(0)$ — coherent

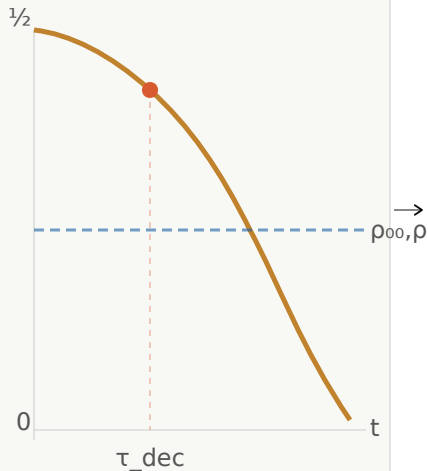


superposition: $|\rho_{01}| = \frac{1}{2}$

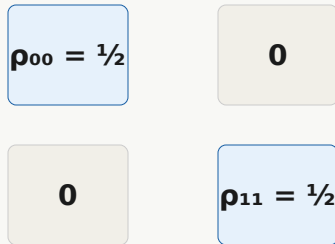
DP Sea scattering:
random phase kicks $\delta\varphi$
 $\gamma = \text{sea_str}^2 E_P/\hbar$

off-diagonal decay

$$|\rho_{01}(t)| = \frac{1}{2} e^{(-2\gamma t)}$$



$\rho(\infty)$ — collapsed



classical mixture
apparent collapse

Nexus: global unitarity
 $\text{Tr}(\rho_{\text{tot}}^2) = 1$ preserved